

Pet Adoption System — Docker Setup Chat

A chronological record of Docker hosting setup, troubleshooting, and final commands.

1. Containerizing the project

User

I wanna host this project to a docker. please do it

Assistant

Added a three-service Docker Compose stack: Oracle XE, Express API, and a production Vite/React frontend served by Nginx. Added Dockerfiles, Nginx reverse proxying for /api, database initialization scripts, environment template, Docker ignore files, and Docker instructions in the README.

Added configuration: docker-compose.yml, backend/Dockerfile, frontend/Dockerfile, frontend/nginx.conf, database/docker/00_prepare_user.sql, and database/docker/01_initialize.sh.

2. Docker command not found

User

docker : The term 'docker' is not recognized as the name of a cmdlet, function, script file, or operable program.

Assistant

The Docker CLI was not installed or not available in PowerShell. Docker Desktop was installed, then WSL was checked and updated.

User

```
wsl --install
A distribution with the supplied name already exists.

wsl --update
The most recent version of Windows Subsystem for Linux is already installed.

docker version
Client and Server: Docker Desktop 4.89.0
```

Assistant

Docker was confirmed to be correctly installed. The existing WSL distribution is normal; no further `wsl --install` command is needed.

3. Compose file not found

User

```
C:\Users\LENOVO>docker compose up --build
no configuration file provided: not found
```

Assistant

Compose was being run from the wrong folder. It must run from the repository root, where `docker-compose.yml` is located.

```
cd /d E:\pet-adoption-system
docker compose up --build
```

4. Oracle port conflict

User

```
ports are not available: exposing port TCP 0.0.0.0:1521 ... bind: Only one usage of each
socket address is normally permitted.
```

Assistant

Port 1521 was already used by a local Oracle installation. The host mapping was changed from `1521:1521` to `1522:1521`. Containers still use internal Oracle port 1521.

5. Frontend port conflict and database procedure fix

User

```
Container pet-adoption-system-backend-1 Healthy
Error response from daemon: ports are not available: exposing port TCP 0.0.0.0:8080 ...
```

Assistant

The API and Oracle initialization completed successfully, but 8080 was occupied. The frontend host port was changed to 8081. Schema verification also exposed an invalid adoption procedure referencing a missing sequence; it was fixed to use SYS_GUID().

6. Final startup instructions

This reset removes only the project's Docker Oracle volume. It is appropriate because it contains only freshly created demo data.

```
cd /d E:\pet-adoption-system
docker compose down -v
docker compose up --build --force-recreate
```

After first-time Oracle initialization finishes, open **http://localhost:8081**.

Demo login: admin / password

For external database tools, use localhost:1522/XEPDB1, user PET_ADMIN, and the password configured in .env.

7. Database connection and adding data

The project is connected to Oracle automatically. The backend container uses database:1521/XEPDB1; database is the Docker Compose service name. The frontend calls the backend through Nginx, so all three services are connected.

Use SQL Developer (recommended): create a new Oracle connection with Host localhost, Port 1522, Service name XEPDB1, Username PET_ADMIN, and the value of APP_USER_PASSWORD in the root .env file.

Use the terminal: replace YOUR_APP_USER_PASSWORD with the value in .env.

```
cd /d E:\pet-adoption-system
docker compose exec database sqlplus PET_ADMIN/YOUR_APP_USER_PASSWORD@//localhost:1521/XEPDB1
```

Inside SQL*Plus or SQL Developer, add data with an INSERT, then persist it with COMMIT. Example:

```
INSERT INTO PERSON (PERSON_ID, FIRST_NAME, LAST_NAME, DATE_OF_BIRTH, GENDER, EMAIL)
VALUES ('P021', 'Amina', 'Rahman', DATE '1998-06-15', 'FEMALE', 'amina@example.com');

COMMIT;

SELECT PERSON_ID, FIRST_NAME, LAST_NAME, EMAIL
FROM PERSON
WHERE PERSON_ID = 'P021';
```

Use a new, unique primary key and respect foreign-key relationships: insert parent rows (for example, PERSON) before dependent rows (for example, PERSON_PHONE). Changes stored in the Docker volume remain after `docker compose down`; they are erased only by `docker compose down -v`.

Useful Docker commands

```
docker compose ps
docker compose logs -f
docker compose logs -f database
docker compose down
```